

Integration

1.1 Integration Respect to a Measure

Theorem 1.1.1 ► Monotone Convergence Theorem

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 \leq f_1 \leq f_2 \leq \dots$ is an increasing sequence of $\mathcal{S}$ -measurable functions. Define $f : X \rightarrow [0, \infty)$ by

$$f(x) = \lim_{k \rightarrow \infty} f_k(x).$$

Then

$$\lim_{k \rightarrow \infty} \int f_k d\mu = \int f d\mu.$$

Proof. f is $\mathcal{S}$ -measurable as the pointwise limitation of $\{f_k\}$. Since $f_k \leq f$ for all k , we have $\int f_k d\mu \leq \int f d\mu$. Thus

$$\lim_{k \rightarrow \infty} \int f_k d\mu \leq \int f d\mu$$

To show the converse, take arbitrary simple function $s = \sum_{j=1}^n c_j \chi_{A_j}$ with standard representation such that $f \geq s$. We fix a $t \in (0, 1)$, and define

$$E_k := \left\{ x \in X : f_k(x) \geq t \sum_{j=1}^n c_j \chi_{A_j}(x) \right\}$$

Then E_k s are measurable and

$$E_1 \subseteq E_2 \subseteq \dots, \quad \bigcup_{k=1}^{\infty} E_k = X$$

Then it follows that $\mu(A_j) = \lim_{k \rightarrow \infty} \mu(A_j \cap E_k)$ for each j . We have

$$\int f_k d\mu \geq \int t \sum_{j=1}^n c_j \chi_{A_j \cap E_k} d\mu \geq t \sum_{j=1}^n c_j \mu(A_j \cap E_k)$$

By taking $k \rightarrow \infty$, we get

$$\lim_{k \rightarrow \infty} \int f_k d\mu \geq t \sum_{j=1}^n c_j \mu(A_j)$$

Taking the supremum of the inequality above over all simple functions s , we get

$$\lim_{k \rightarrow \infty} \int f_k d\mu \geq t \int f d\mu$$

Finally, taking the limit of t increasing to 1 shows that

$$\lim_{k \rightarrow \infty} \int f_k d\mu \geq \int f d\mu$$

□

Lemma 1.1.2 ► Fatou's lemma

If $\{f_k\}_{k=1}^{\infty}$ is a sequence of nonnegative μ -measurable functions, then

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu.$$

Proof. Let $g_k(x) = \inf_{n \geq k} f_n(x)$. Then $0 \leq g_1 \leq g_2 \leq \dots$ is an increasing sequence of $\mathcal{S}$ -measurable functions. Then

$$g(x) = \liminf_{k \rightarrow \infty} f_k$$

It follows from Monotone Convergence Theorem that

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu = \lim_{k \rightarrow \infty} \int_X g_k d\mu = \lim_{k \rightarrow \infty} \int_X g_k d\mu$$

Since for each g_k , we have

$$g_k(x) = \inf_{n \geq k} f_n(x) \leq f_k(x)$$

then

$$\int_X g_k d\mu \leq \int_X f_k d\mu, \quad \forall k \geq 1$$

By taking the $\liminf$ relative to k both sides, we have

$$\lim_{k \rightarrow \infty} \int_X g_k d\mu = \liminf_{k \rightarrow \infty} \int_X g_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu.$$

Thus

$$\int_X \liminf_{k \rightarrow \infty} f_k d\mu \leq \liminf_{k \rightarrow \infty} \int_X f_k d\mu$$

□

1.2 Limits of Integrations

1.2.1 Bounded Convergence Theorem

Proposition 1.2.1 ► bounding an integral

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $E \in \mathcal{S}$, and $f : X \rightarrow [-\infty, \infty]$ is a function such that $\int_E f d\mu$ is defined. Then

$$\left| \int_E f d\mu \right| \leq \mu(E) \sup_E |f|.$$

Proof. Let $c = \sup_E |f|$, then

$$\begin{aligned}
 \left| \int_E f \, d\mu \right| &= \left| \int \chi_E f \, d\mu \right| \\
 &\leq \int |\chi_E f| \, d\mu \\
 &\leq \int \chi_E c \, d\mu \\
 &= \mu(E) c = \mu(E) \sup_E |f|
 \end{aligned}$$

□

1.2.2 Dominated Convergence Theorem

Proposition 1.2.2 ► integrals on small sets are small

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $g : X \rightarrow [0, \infty]$ is $\mathcal{S}$ -measurable, and $\int g \, d\mu < \infty$. Then for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\int_B g \, d\mu < \varepsilon$$

for every set $B \in \mathcal{S}$ such that $\mu(B) < \delta$.

Remark.

- $g \in L^1(\mu)$ induces a finite measure

$$\nu(B) = \int_B g \, d\mu, \quad B \in \mathcal{S}$$

- For each $\varepsilon > 0$, there exists $\delta > 0$ such that $\nu(B) < \varepsilon$ whenever $\mu(B) < \delta$.

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}^n} |f(x+h) - f(x)| \, dx = 0$$

Proof. Since $\int g \, d\mu < \infty$, for each $\varepsilon > 0$, there exists a simple function

$h : X \rightarrow [0, \infty]$ such that $h \leq g$,

$$0 \leq \int g \, d\mu - \int h \, d\mu < \frac{\varepsilon}{2}$$

Suppose that $c = \sup_X h$, which is finite. Then for $\delta = \frac{\varepsilon}{2c}$, we have for each $B \in \mathcal{S}$ such that $\mu(B) < \delta$, there is

$$\int_B h \, d\mu < \mu(B) c < \frac{\varepsilon}{2}$$

$$\int_B g \, d\mu \leq \int_B h \, d\mu + \frac{\varepsilon}{2} < \varepsilon$$

□

Proposition 1.2.3 ▶ integrable functions live mostly on sets of finite measure

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $g : X \rightarrow [0, \infty]$ is $\mathcal{S}$ -measurable, and $\int g \, d\mu < \infty$. Then for every $\varepsilon > 0$, there exists $E \in \mathcal{S}$ such that $\mu(E) < \infty$ and

$$\int_{X \setminus E} g \, d\mu < \varepsilon.$$

Proof. Let $h : X \rightarrow [0, \infty]$ be a simple function such that $0 \leq h \leq g$,

$$\int g \, d\mu - \int h \, d\mu < \frac{\varepsilon}{2}$$

Define

$$E = h^{-1}((0, \infty))$$

It follows from the definition of the simple function and its integrability that $\mu(E) < \infty$. Furthermore,

$$\int_{X \setminus E} h \, d\mu = \int_{X \setminus E} 0 \cdot d\mu = 0$$

Thus

$$\int_{X \setminus E} g \, d\mu \leq \frac{\varepsilon}{2} + \int_{X \setminus E} h \, d\mu < \varepsilon$$

□

Theorem 1.2.4 ► Dominated Convergence Theorem

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $f : X \rightarrow [-\infty, \infty]$ is $\mathcal{S}$ -measurable, and $f_1, f_2, \dots$ are $\mathcal{S}$ -measurable functions from X to $[-\infty, \infty]$ such that

$$\lim_{k \rightarrow \infty} f_k(x) = f(x)$$

for almost every $x \in X$. If there exists an $\mathcal{S}$ -measurable function $g : X \rightarrow [0, \infty]$ such that

$$\int g \, d\mu < \infty \quad \text{and} \quad |f_k(x)| \leq g(x)$$

for every $k \in \mathbb{Z}^+$ and almost every $x \in X$, then

$$\lim_{k \rightarrow \infty} \int f_k \, d\mu = \int f \, d\mu.$$

Proof sketch.

将积分差先控制在 g 几乎落在其上的有限测度集 E , 再将 E 上的积分分别控制在一个小测度集 B_ε 和使得 f_k 一致收敛的大测度集 $E \setminus B_\varepsilon$ 上.

Proof. For each $E \in \mathcal{S}$, and $k \in \mathbb{Z}^+$,

$$\begin{aligned} \left| \int f_k \, d\mu - \int f \, d\mu \right| &\leq \int_E |f_k - f| \, d\mu + \left| \int_{X \setminus E} f_k \, d\mu \right| + \left| \int_{X \setminus E} f \, d\mu \right| \\ &\leq \int_E |f_k - f| \, d\mu + 2 \int_{X \setminus E} g \, d\mu \end{aligned}$$

From proposition 1.2.3, for every $\varepsilon > 0$, there exists $E_\varepsilon \in \mathcal{S}$ such that $\mu(E_\varepsilon) < \infty$ and

$$\int_{X \setminus E_\varepsilon} g \, d\mu < \frac{\varepsilon}{4}$$

From proposition 1.2.2, there exists $\delta > 0$ such that for each $F \in \mathcal{S}$, $\mu(F) < \delta$,

we have

$$\int_F g \, d\mu < \frac{\varepsilon}{8}$$

The Egorov's theorem implies that there exists measurable set $B_\varepsilon \in \mathcal{S}$ with $\mu(E_\varepsilon \setminus B_\varepsilon) < \delta$ such that f_k converges uniformly to f on B_ε . Then we have

$$\lim_{k \rightarrow \infty} \int_{B_\varepsilon} |f_k - f| \, d\mu \leq \lim_{k \rightarrow \infty} \mu(B_\varepsilon) \sup |f_k - f| = 0$$

Then there exists N such that

$$\int_{B_\varepsilon} |f_k - f| \, d\mu < \frac{\varepsilon}{4}, \quad k \geq N$$

Furthermore, for each k we have

$$\int_{E_\varepsilon \setminus B_\varepsilon} |f_k - f| \, d\mu \leq 2 \int_{E_\varepsilon \setminus B_\varepsilon} g \, d\mu < \frac{\varepsilon}{4}$$

Therefore

$$\int_{E_\varepsilon} |f_k - f| \, d\mu \leq \int_{E_\varepsilon \setminus B_\varepsilon} |f_k - f| \, d\mu + \int_{B_\varepsilon} |f_k - f| \, d\mu < \frac{\varepsilon}{2}, \quad k \geq N$$

then it follows that

$$\left| \int f_k \, d\mu - \int f \, d\mu \right| \leq \int_{E_\varepsilon} |f_k - f| \, d\mu + 2 \int_{X \setminus E_\varepsilon} g \, d\mu < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad k \geq N$$

That is

$$\lim_{k \rightarrow \infty} \int f_k \, d\mu = \int f \, d\mu$$

□

1.3 Integrable Functions and Continuous Functions

Definition 1.3.1

Denote $C_c(\mathbb{R}^n)$ by the collection of compactly supported continuous functions.

Theorem 1.3.2

Let E be a measurable set. $C_c(\mathbb{R}^n)$ is dense in $L(E)$. That is if $f \in L(E)$, then for each $\varepsilon > 0$, there exists $g \in C_c(\mathbb{R}^n)$ such that

$$\int_E |f - g| \, d\mu < \varepsilon$$